

HDOC - Right-Aligned Patterns

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Problem Statement

Write a menu-driven C program using a switch statement to print right-aligned patterns selected by the user. Use nested loops without arrays. Implement each pattern in a separate parameterless function with no return value.

Patterns

1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle

Algorithm

1. Start the program.
2. Display the menu containing the three pattern choices.
3. Read the user's choice.
4. Use switch-case to call the corresponding parameterless void function.
5. Inside the selected function, input the number of rows n.
6. Use the outer loop to control rows.
7. Use the first inner loop to print leading spaces for right alignment.
8. Use another inner loop to print stars or natural numbers.
9. Print a newline after each row.
10. Return to main() and terminate the program.

Test-Case Table

Test	Choice	n	Expected Result	Actual Result	Result
TC1	1	4	Increasing star triangle, 4 rows	Same as expected	Pass
TC2	2	4	Natural-number triangle, 4 rows	Same as expected	Pass
TC3	3	4	Decreasing star triangle, 4 rows	Same as expected	Pass
TC4	1	1	Single *	Same as expected	Pass
TC5	2	3	Rows: 1; 1 2; 1 2 3	Same as expected	Pass

C Program

```
#include <stdio.h>

void increasingTriangle()
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++)
    {
        for (j = 1; j <= n - i; j++)
            printf(" ");
        for (j = 1; j <= i; j++)
            printf("* ");
        printf("\n");
    }
}

void increasingNumberTriangle()
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++)
    {
        for (j = 1; j <= n - i; j++)
            printf(" ");
        for (j = 1; j <= i; j++)
            printf("%d ", j);
        printf("\n");
    }
}

void decreasingTriangle()
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = n; i >= 1; i--)
    {
        for (j = 1; j <= n - i; j++)
            printf(" ");
        for (j = 1; j <= i; j++)
            printf("* ");
        printf("\n");
    }
}

int main()
{
    int choice;

    printf("RIGHT-ALIGNED PATTERNS\n");
    printf("1. Right-aligned increasing triangle\n");
    printf("2. Right-aligned increasing number triangle\n");
    printf("3. Right-aligned decreasing triangle\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            increasingTriangle();
            break;
        case 2:
            increasingNumberTriangle();
            break;
        case 3:
            decreasingTriangle();
            break;
    }
}
```

```

        break;
    case 3:
        decreasingTriangle();
        break;
    default:
        printf("Invalid choice\n");
    }
    return 0;
}

```

Sample Output for n = 4

1. Increasing triangle

```

    *
   * *
  * * *
 * * * *

```

2. Increasing number triangle

```

    1
   1 2
  1 2 3
 1 2 3 4

```

3. Decreasing triangle

```

* * * *
 * * *
  * *
   *

```

Header file used: <stdio.h>

Library functions used: printf() and scanf()

Execution / Actual Output

```
Last login: Mon Sep 21 23:13:17 on ttys000
dhruvpandey@DHRUVs-MacBook-Air ~ % nano day27.c
dhruvpandey@DHRUVs-MacBook-Air ~ % clang day27.c -oday27
dhruvpandey@DHRUVs-MacBook-Air ~ % ./day27
RIGHT-ALIGNED PATTERNS
1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
Enter your choice: 1
Enter number of rows: 4
*
**
***
****
dhruvpandey@DHRUVs-MacBook-Air ~ % ./day27
RIGHT-ALIGNED PATTERNS
1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
Enter your choice: 2
Enter number of rows: 4
  1
 1 2
1 2 3
1 2 3 4
dhruvpandey@DHRUVs-MacBook-Air ~ % ./day27
RIGHT-ALIGNED PATTERNS
1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
Enter your choice: 3
Enter number of rows: 4
****
***
**
*
dhruvpandey@DHRUVs-MacBook-Air ~ % ./day27
RIGHT-ALIGNED PATTERNS
1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
Enter your choice: 1
Enter number of rows: 1
*
dhruvpandey@DHRUVs-MacBook-Air ~ % ./day27
RIGHT-ALIGNED PATTERNS
1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
Enter your choice: 2
Enter number of rows: 3
  1
 1 2
1 2 3
dhruvpandey@DHRUVs-MacBook-Air ~ %
```